

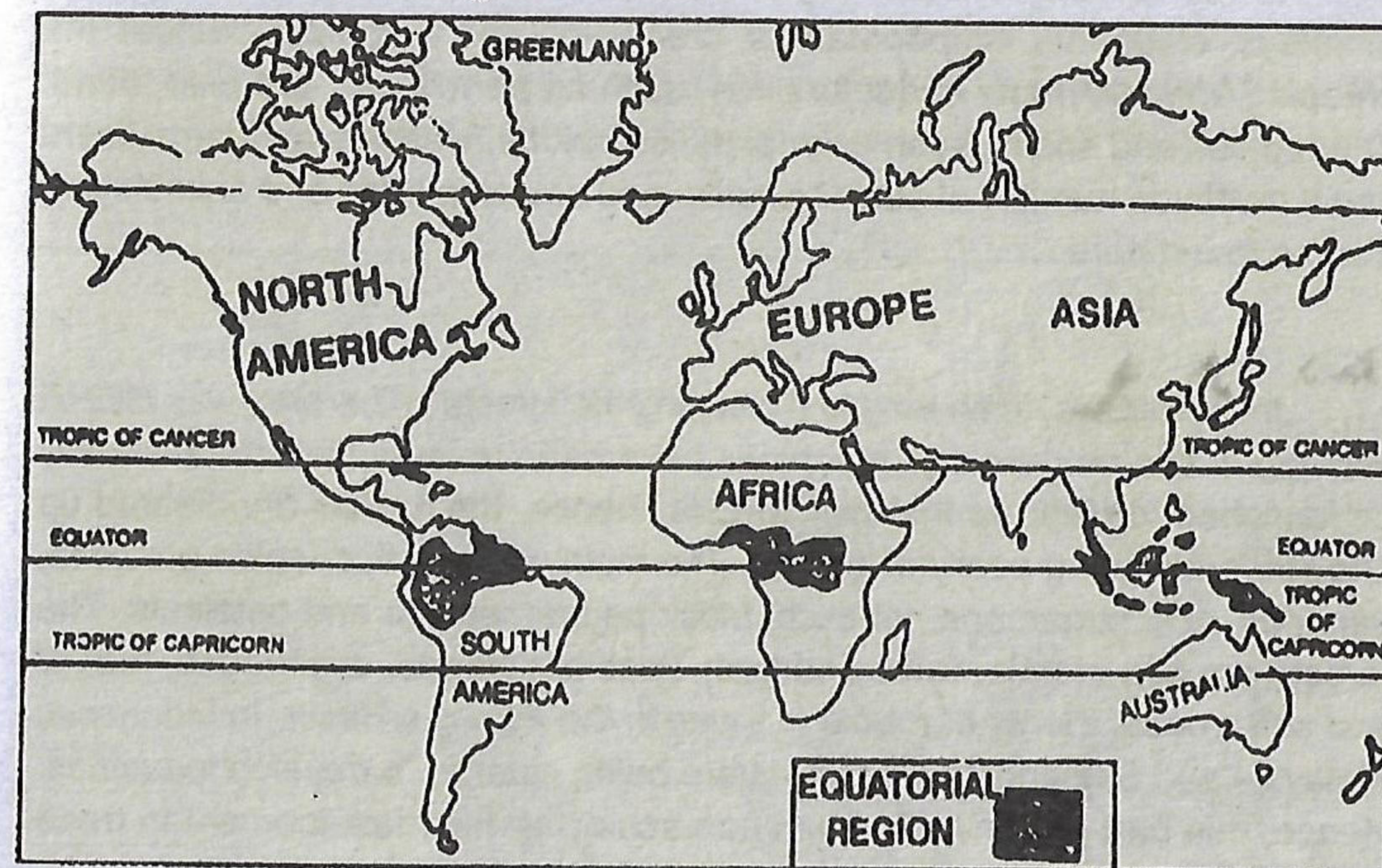
Major Climatic Regions of the World

In order to acquire better knowledge about the world, the globe has been distributed among a number of imaginary climate regions or the belts. Such distribution has been made on the basis of the distance from the Equator and the Poles. The Equator is the imaginary line that passes through the centre of the Earth horizontally. It falls on the zero degree. All the year round, the Sun rays fall on it in vertical position (i.e. 90°), and due to this reason, the countries situated on, or near the Equator are very hot. Also due to heat, greater quantity of water is evaporated as compared to other places. This is also a region of high rainfall. The Tropic of Cancer lies in the north. At 23° from the Equator, and the Tropic of Capricorn at $23\frac{1}{2}^\circ$ south (of equator). These tropics are also imaginary lines. The entire space between the Tropic of Cancer and Tropic of Capricorn is called the 'Hot Belt'. As compared to other belts, this belt is richer in varieties of the plants and animals. Different kinds of flowers also grow in this area. The countries situated in this region are very hot. However, as we go further away from the equator, the intensity of heat in the area decreases. From $23\frac{1}{2}^\circ$ to $66\frac{1}{2}^\circ$ N and $23\frac{1}{2}^\circ$ S to $66\frac{1}{2}^\circ$ S, the belt is called (Temperate Zone) Mediterranean belt. The countries lying in between this belt are temperate neither very cold nor very hot. Most of the European countries lie within this region or the belt. But by getting closer to the poles, the intensity of cold gradually increases. From $66\frac{1}{2}^\circ$ north and south, up to the poles, it is called the "Cold belt". Due to severe cold this belt generally remains covered with snow. In order to collect more information, these belts have further been classified into many parts from the point of view of climate. These parts are called the "Climatic regions" or the "Natural regions". The major climatic regions are:-

1. The Equatorial Region.
2. The Monsoon Region.
3. The Mediterranean Region.
4. The Temperate Grasslands or Steppes Region.
5. The Hot Desert Region.
6. The Tundra Region.

1. The Equatorial Region

This region comprises lands situated between 5° North and 5° South latitude with the Equator running right through the middle of it. It consists of Malaysia, Indonesia, Colombia, Amazon Basin, Guinea, Congo Basin and southeastern coastal region of Africa.



Climature

Being closer to the Equator, this region is comparatively more exposed to the evaporative activity. Hence, the winds become humid. When the humid winds rise up due to the low pressure, they become cold and cause rains. Here, it rains almost regularly after three O'clock in the daytime. Since, all the year round the sun rays fall vertically (at 90°) this region is very hot. In the Amazon Basin of South America, the average annual rainfall is about two thousand millimeters. Due to the excessive rainfall and heat, the climate of this region is not good for health.

Vegetation

Excessive rains have resulted in dense vegetation in this region. Evergreen forests are found everywhere. Dense growth of the forests obstructs the light of the sun from reaching the ground below. The resultant darkness, dampness

and steaming vapour have enabled mosquitoes and other parasites to grow in swarms.

Animal Life

In the forests of these regions different animals are found such as monkeys, elephants, hippopotamus, crocodiles and serpents. Besides the creeping animals many beautiful birds, such as parrots etc. are also found. The reptiles and snails, such as snakes, scorpions, weasels and large lizard and a multitude number of other beautiful and amazing birds and animals can also be found here.

Agriculture

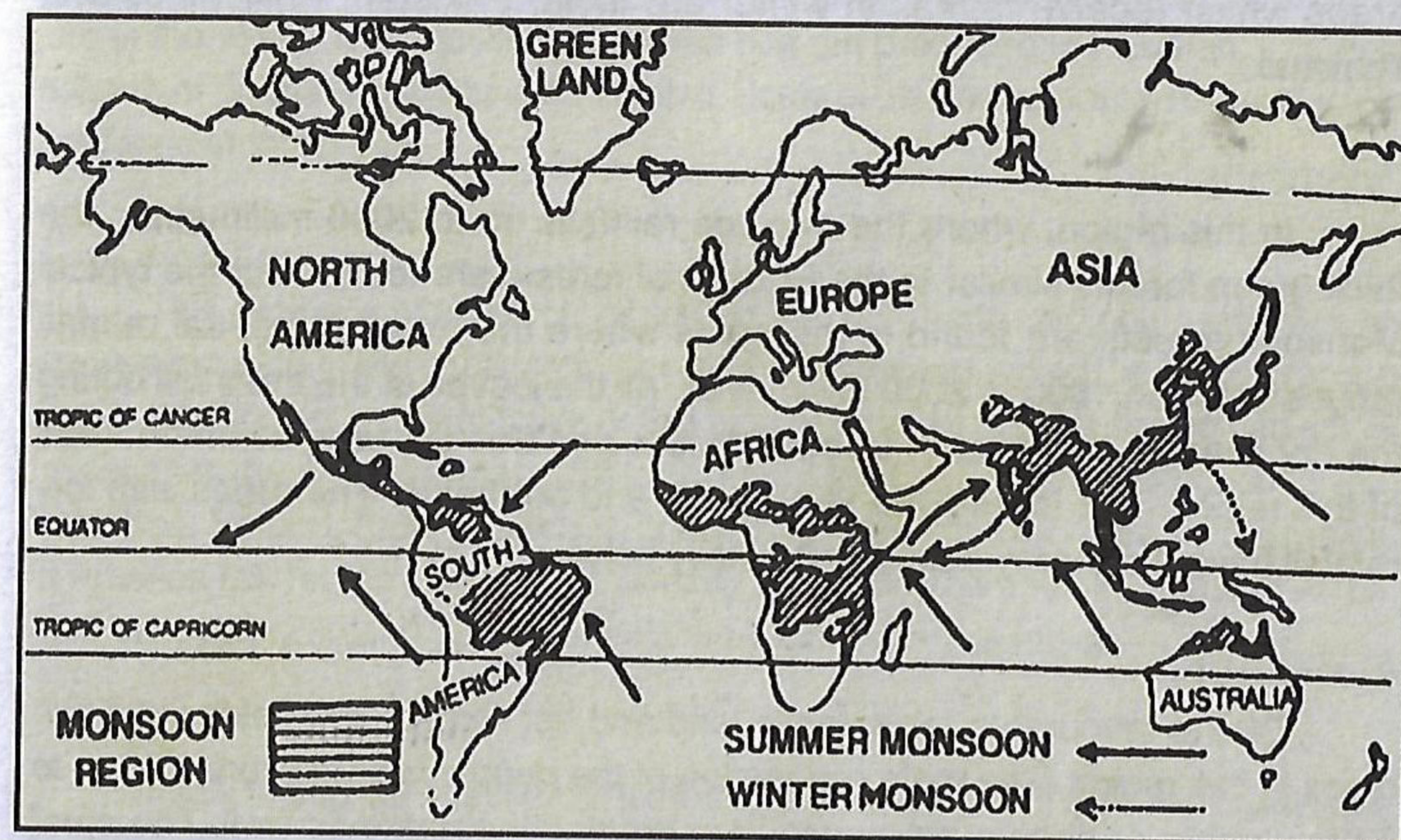
In olden days, lumbering and cleaning the forests in this area was difficult task but in this modern age machines have been invented which can easily be launched to clear up the thick forests. Hence, the forests are cleared up and lands are being easily cultivated. The main crops of this region are coca, bananas, rice, sugarcane, tobacco, black pepper, spices and betelnuts. The forests are rich in nuts, fruits and gums, cabinet woods, dye-woods, rubber and soft woods. Plenty of rubber is found in the Amazon Basin. In Indonesia, Malaysia and Singapore, the forests are being cleared to develop industries. Hence, this part of the world has been achieving fast development in trade and industry. Various types of products such as rubber sugar, spices, cocoa, Saboo-rice, etc. are exported from this region to other parts of the world. Generally, the people of the Equatorial region are backward. Due to the difficulties of transport and communication system the trade of this region is not well organized and well developed. The main occupation of the people of this area is hunting and cutting the wood whenever the conditions are favourable. Wherever there are forests, the general conditions of this region are being changed fast now.

Minerals

This region is generally poor in the mineral resources. Some countries are considerably important in this respect. For example, in Malaysia and Indonesia nickle and coal are found in greater quantities. In Indonesia, oil has also been found. Huge quantities of copper are found in Congo. At some places zinc, silver and gold are also found.

2. Monsoon Region

The monsoon-region lies on both sides of the Equator between 5 to 30 degrees, on the eastern sides of the continents. The following countries are situated within this belt. In Asia, Pakistan, Bangladesh, Myanmar, Vietnam, Kampuchea, Laos, and the Islands of Philippines. In North America, the eastern coast of Brazil, Mexico and Venezuela in Central America, in Africa, the countries of the East Africa and Malagasy Island. And the Northern part of Australia.



Climate

In the summer season, the winds blow from sea towards the lands and contain sufficient vapour and cause enough rainfall in mountainous areas. As the direction of these winds is quite the reverse in winter i.e. they blow from land towards the sea hence they are dry and cause no or very little rain in winter in this region. The climate of this region is hot and humid during summer but dry during winter. Besides the summer and the winter, there is also the rainy season. The summers are extremely hot, but the winters are moderate. Generally, the rainy season starts from the month of June and continues up to the month of October. These rains depend upon the directions of the winds,

the pressure of the winds, and the distance of the land from the sea. In Pakistan, it rains a lot in the northern mountainous area due to the monsoon winds from the Arabian Sea. But the average rainfall decreases gradually towards the south.

Bangladesh and Assam are located on the route of northeasterly monsoons of the Bay of Bengal hence they get plenty of rainfall. Some countries of monsoon region receive rains in winter in addition to the summer rainfall. Areas which receive rainfall in winter are India, Pakistan, Philippines and Thailand.

Vegetation

In this region, where the average rainfalls up to 2000 millimetres, the thick green forests similar to the Equatorial forests are found. But the typical Monsoon forests are found in the areas where the average annual rainfall ranges between 1000 to 2000 millimetres. All the leaves of the trees fall during the dry season. Mahogany, Deodar, Neem and Sall are very common trees of this region. The forests are very useful and profitable. The areas with low rainfall have the thorny plants and grassy fields.

Agriculture

Due to favourable rains, fertile land and hot climate, the fields yield rich crops in this region. The main occupation of the people is agriculture, but quite a big number of people are engaged in grazing the cattle as well. The chief crops of the area are rice, jute and sugarcane, while some seeds, pulses are also grown. Rice is grown in deltas. Bananas, coconut, guava, mango and pineapple are the major fruits of this region.

Animal Life

Since this region is covered with thick forests, many animals are found in them. The common animals are tiger, elephant and deer. Among the domestic animals, there are cow, buffallow, horse, goat, camel and sheep. Good quality leather is made from the hides of these animals.

Minerals

This region is quite rich in mineral resources. Different kinds of minerals such as silver, iron, lead, coal, tin, mica and oil have been found. At some places gold is also discovered.

Some areas of the Monsoon Region are very developed and their means of transportation are also sufficiently developed. Such areas are the most densely populated parts of the world. Some facts are worth mentioning regarding the Muslim countries lying in the Monsoon Region. Bangladesh produces largest quantity of jute in the world. Best quality cotton and rice are produced in Pakistan. The best network of canals is also found in Pakistan. Some of the best specimen of architecture are found in this region.

3. Mediterranean Region

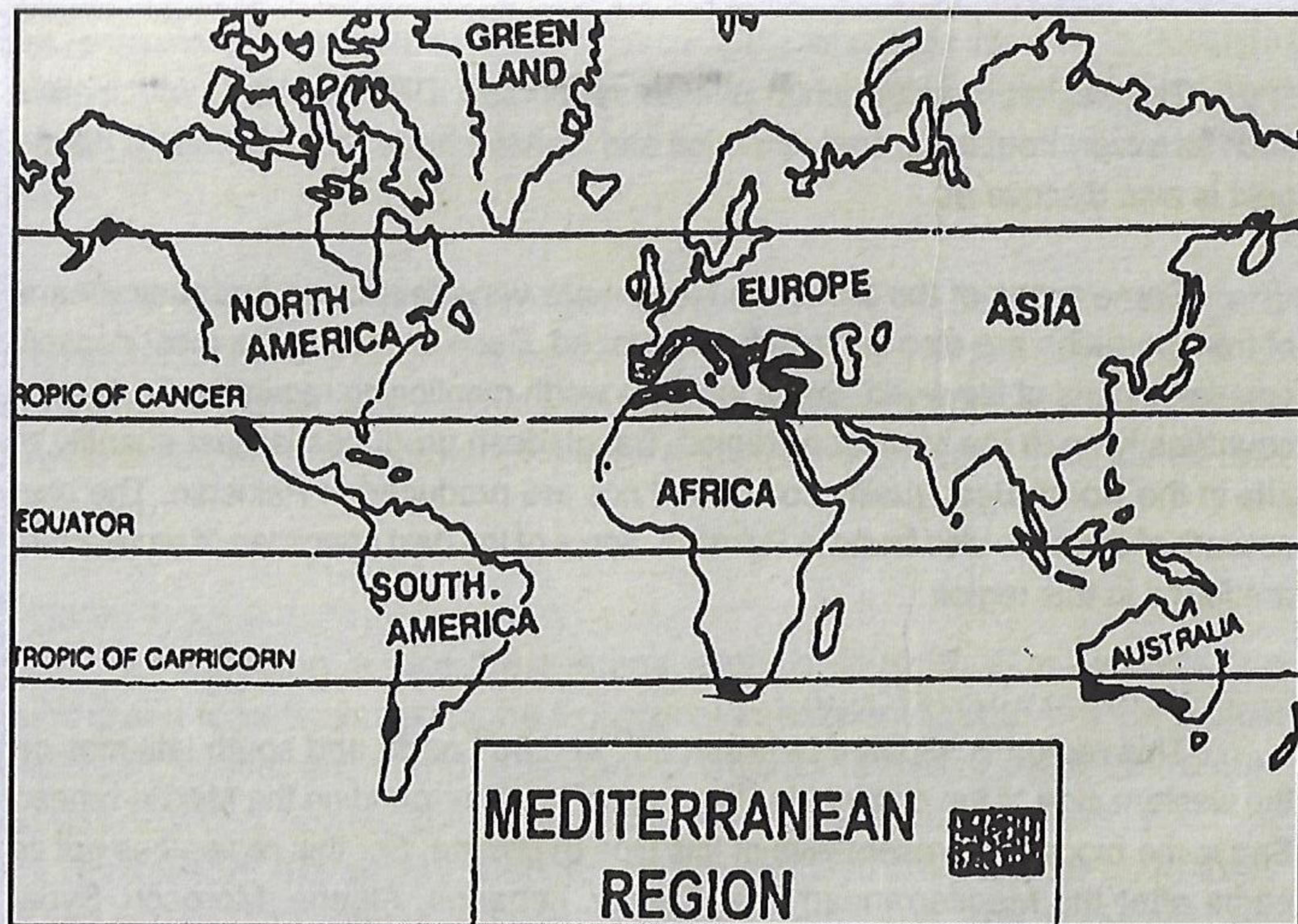
This region is situated between 30° and 45° north and south latitudes on the western side of the continents. The tract of land surrounding the Mediterranean Sea is the biggest representative of this type of climate. So, this region has got its name after the Mediterranean Sea. Turkey, Lebanon, Algeria, Morocco, Syria, Palestine, France, Italy, Spain and the Balkan states are included in this region. In America this region comprises California and Chile. The southern parts of Africa, Australia, New Zealand and Tasmania are included in this region.

Climate

In this region the summer season is short and dry and the winter is pleasant. The rain falls in the winter season with an annual average ranging between 400 to 500 millimetres. The mountainous areas exposed to the mild winds are also exposed to the torrential rains. The southern part of the Mediterranean Sea however, gets comparatively less rainfall. Hence, this area further enters into the semi-desert region.

Vegetation

There are shorter trees, but their roots are spread widely into the land. The leaves of these trees are small glistening and thick, so that their moisture does not come out easily due to the Sun. Among the most common trees are olive, fig, cork, oak, deodar and sofaida. This region is famous for the production of citrus fruits all over the world such as grapes, oranges, lemons, figs, apricots mulberries, almonds



and nuts which are grown here in abundance. The areas, which have rainfall or have irrigation systems, produce wheat cotton, barley and millet (bajra). Rice is cultivated along the banks of the rivers. The most important occupation of the people of this region is cultivation of the fruits. Sheep and goats are also reared.

Minerals

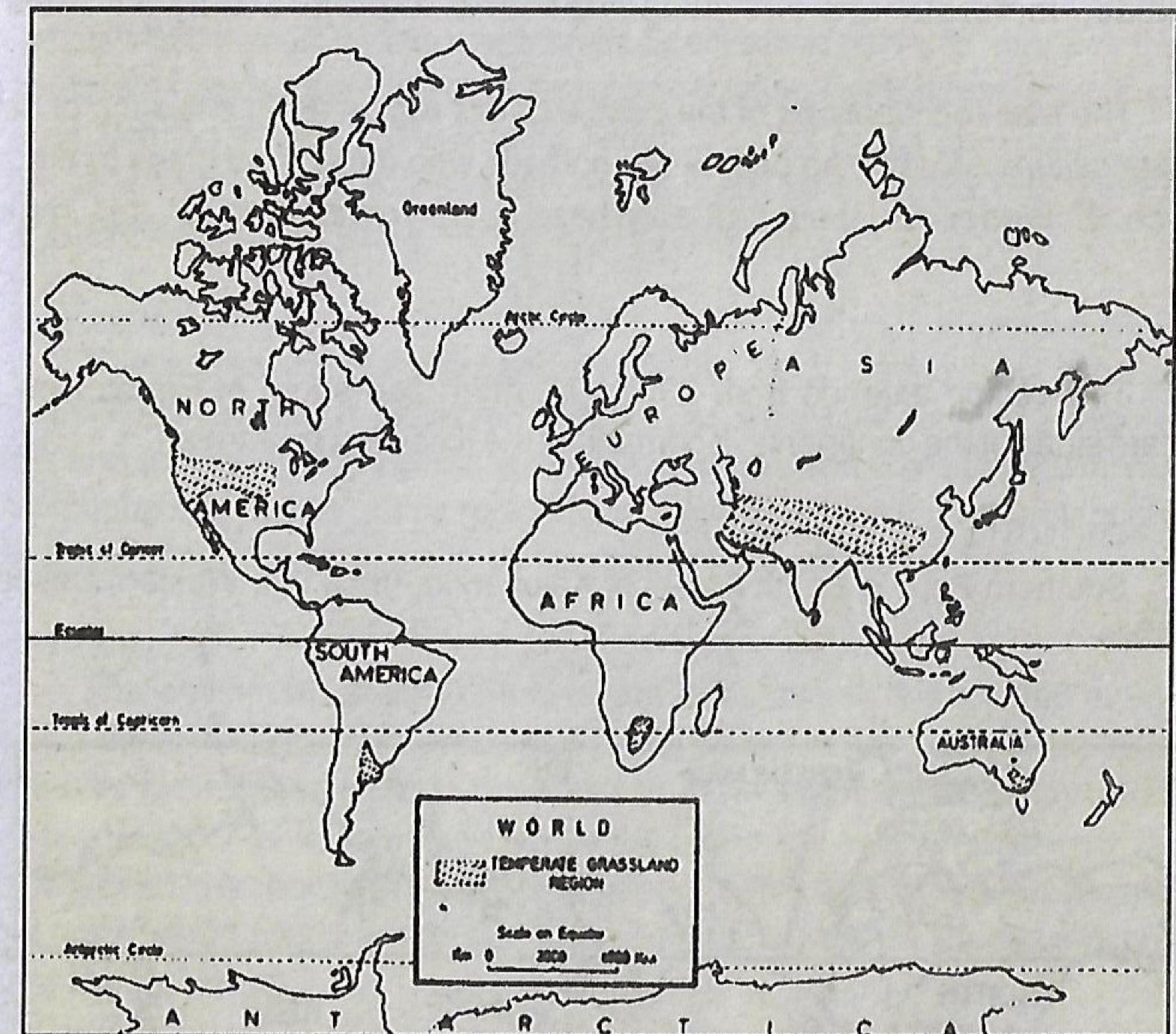
Iron is found in the coastal areas of this region. In Spain apart from iron, mercury is also found. In other areas copper, zinc and lead is found. In fact, this region is deficient in the mineral resources.

4. Temperate Grasslands or Steppes Region

This region lies in the central parts of the continents. In the northern hemisphere this climatic belt includes southern Siberia, central European Russia, Germany, Poland, Sweden, Hungary and Romania. The northern states of the U.S.A. and southern Canada are also situated in this region. In the southern hemisphere the areas are Argentina, Wolds in South Africa and the Dones area of Australia.

Climate

This region has severe summers and winters. Hot winds blow in summer and cold winds blow in winter. Rainfall is scarce being between 250 and 500 mms., which generally falls in summer.



Vegetation

Due to the low rainfall grass is the only natural growth in this region. No tree could be seen in the vast expanses of grass. In the rainy season it is green all around. And due to the abundance of grass there are many grazing fields. There are also many grass eating and fast-running animals such as antelope, rabbit, goat, and sheep. Due to the recent development in the mechanized farming and the facilities of irrigation this area has proved good for cultivation. The crops that easily grow in this region are wheat, oats and mustard. The famous wheat producing areas of America and Canada are also situated in this region.

Mineral Resources

In this region no progress has been achieved in the field of mineral resources. In North America some minerals like oil, coal copper, mica, manganese and gold have been explored. Kuwait and Iraq are famous for oil. Due to the lack of transport facilities, minerals are not excavated and exploited fully in Siberia.

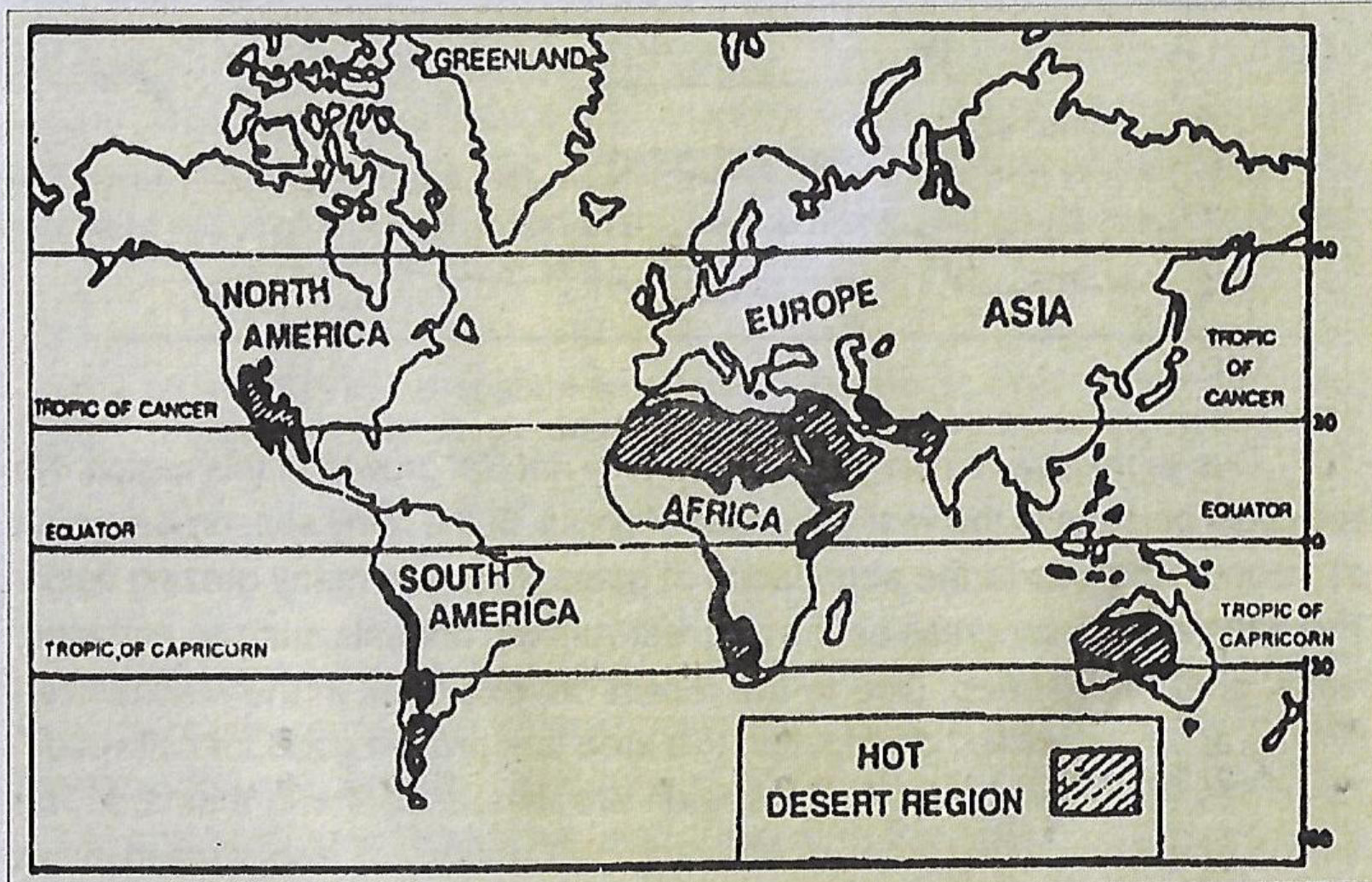
The main occupations of the people of this region are the rearing of cattle and agriculture. Most of the people are nomads who move from place to place in search of fodder for their cattle. Wheat is the chief crop of this region.

5. Hot Desert Region

This region extends from 20° to 30° north and south of Equator on the western sides of the continents. It comprises the following countries:-

(i) In Northern Hemisphere

Southern Peru in South America, Kalahari in Africa and Western Desert in Australia.



Climate

This region contains the hottest areas of the world. At certain places in this region the temperature rises up to 51.5° Centigrade. At Azizia a place in Africa, the temperature has been recorded upto 58° Centigrade. Winters are not so cold but there is a vast difference of temperature in between days and night. The rainfall is usually less than 250 mms. per annum. Some areas get rains after 3 or 4 years. The winds, blowing from the east, become dry while reaching the western coast. Winds, coming from cold areas to hot latitudes do not bring rain. There is no rainy season at all in the vast tracts of this region. Due to the scarcity of rain and water there is no agriculture worth mentioning in this region.

Vegetation

These areas have the hot and dry climate. Such climate is not conducive to the growth of vegetation. However, in the places where the water is available from wells and springs, the fruit gardens have been grown and some cultivation is done. The natural vegetation of this region is confined to a variety of thorny bushes. A few palms grow in the oases. In some places dates are also grown.

Animal Life

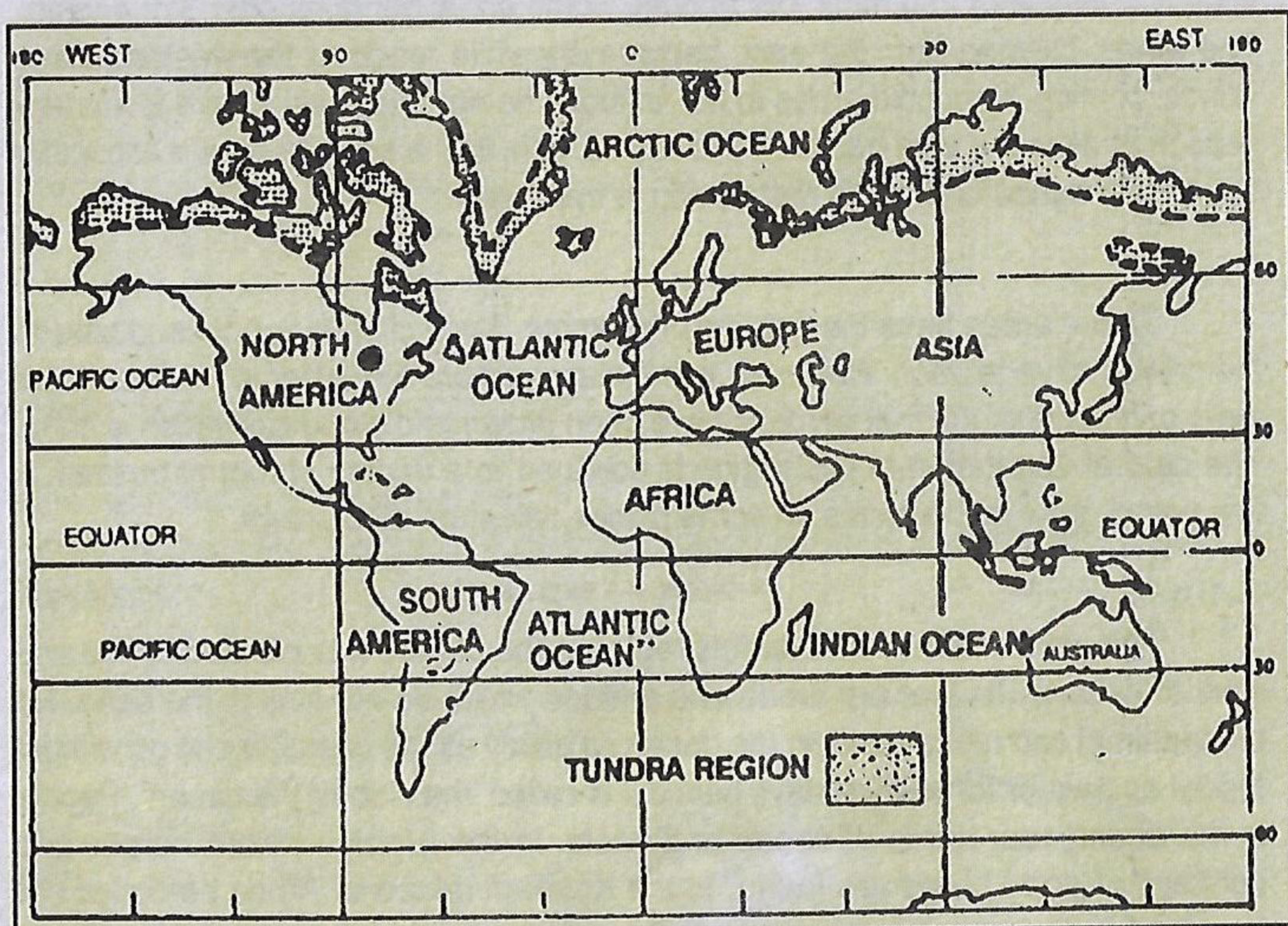
The well-known animal of this region is the camel. It is created to live and work in deserts. Its feet are broad and padded which do not sink in the sand. No other animal can move about in the desert so easily as the camel. It can go without fodder and water for several days hence it is called 'the ship of the desert'. People in the desert rear herds of sheep and goats. In the Arabian desert horses and donkeys of good breed are found. In the Kalahari desert of Africa ostriches are found in big swarms.

Minerals

Some areas of this region have few minerals. In Chile (South America) a large quantity of nitrate is found. In Colorado (North America) silver is mined and in Western Australia, gold is extracted from the mines. In Southern California gold, silver and copper are mined. Oil has been explored in Saudi Arabia, Iraq, Kuwait and Bahrain. Due to the scarcity of food and water, the population is thin in the desert areas. Most of the people lead a nomadic life. They keep camels, sheep, goats and horses. But wherever water is available maize, jowar and millet are cultivated. The good quality of cotton is grown in Egypt. No proper transport facilities are available here. Generally, the camels are used for traveling from one meadow to another.

6. Tundra Region

This region is spread from the Arctic circle to the frozen parts of the Arctic Ocean. This part consists of the extreme northern areas of Asia, Europe and America. It is also called the "Frozen Desert". Like the desert areas this region is also uncultivable due to the extreme cold.



Climate

The climate of the region is extremely cold. The winters are longer than the summers. For twelve months the whole area is covered with snow. The snow melts in the short cool summer but the seas often remain frozen. The melted snow-water spreads over land and changes it into marshes.

Vegetation

Due to the extreme cold, the vegetation is very rare. Some thorny bushes and small plants grow in summer. Besides, some short evergreen trees can also be seen.

Animal Life

Reindeer is the most famous animal of this region. It resembles an antelope. According to the special physical conditions of this region the reindeer is a very

useful animal. The inhabitants of this region also eat the meat of reindeer and make dresses from its hides. Its horns and bones are used for making some tools and hooks for catching the fish. This animal is also used for moving a typical type of carriage on the snow which is called 'Sledge'. Sledges are the only source of transport in this region. Besides reindeer, dog is also a useful animal in this region. They also use the hide of the dog. Some animals such as seals, etc. are also habitats of this region. The skin of seal is used for making shoes and boats.

Minerals

So far this region has remained unexploited for its mineral resources. It is expected that it is rich in minerals. The mining of different types of minerals are on its way. But the pace of exploration work is very slow due to the climatic conditions and the lack of suitable transport facilities.

The people of this region are called ESKIMOS. The word "Eskimos" means "the eaters of raw meat". The main occupation of the people in this region is wood cutting, paper making, collecting turpentine from the jungles, catching fish and hunting the animals having remarkable big fur. The trade of the fur from these animals is very common. These fur-hides are exported to other countries in exchange for other items of their needs.

Exercise

A. Answer the following questions:-

1. What is the difference between "Climatic Region" and "Climatic Belt"?
2. What is a "Climatic Region"?
3. While distributing the Earth in different climatic regions what factors are taken into consideration?
4. What are the different climatic regions of the world?
5. What do you know about the "Equatorial Region"?
6. What are the effects of the winter rainfall in the region of Mediterranean Sea?
7. What are the salient features of the Tundra Region?
8. What are the reasons for the abundant growth of rice in the Monsoon Region?

B) Choose the correct word, and fill in the blanks:-

1. Eskimos use the hides of Seal for making _____.
(Utensils, boats)
2. The _____ are found in excess in the temperate Grassland or Steppes Region.
(Rabbits, Lions)
3. _____ is situated in the Equatorial Region.
(France, Indonesia)

Practical Work

1. Draw a map of the world and show all the main climatic regions in it.
2. Collect the pictures of the houses, built by the people living in different parts of the world.

Human Resources and Professions

Human beings are constantly engaged in different activities. Most important being those by which they procure different things of their need. As civilization dawned these activities grew in number and variety. Man and his wants are the moving force of all economic activities. As such, human resource has assumed great importance in the present day world.

People all around the world earn their living by adopting some or the other profession. These occupations are in accordance with the physical features, climate, production and ways of living of the people of that region. In green and fertile areas where water is available in abundance, cultivation is done. Places where minerals are found, people adopt the profession of mining. In some areas cattle are reared and in some areas fishing is carried out. In Industrial cities labourers work in mills and factories. Industrialists market their manufactured goods and engage themselves in trade and commerce. Banking and Finance are auxiliaries to trade and are also important. The occupations of urban areas are more diversified and require a lot of mental ability. No any occupation is less important than any other. Every occupation requires appropriate education and skills. We should respect all occupations. The details of some major occupations are given as under:

Cattle Rearing

Some regions of the world are dry. There is no rainfall in these regions and due to shortage of water there is no system of irrigation. Therefore, cultivation does not take place. As grass grows in abundance, these areas serve as pasture lands for cattle. From cattle milk, butter, ghee (clarified butter fat), and meat is obtained which is the main food item of the majority of the people. The profession of cattle rearing is mostly adopted in hot and temperate climatic regions. Farms are made in different regions where cattle in large quantity are bred. These farms are near the cities or the industrial centres so that market is available for the sale of milk, meal and butter, etc. In the areas where there is water-storage facility but the climate is extreme, cattle rearing is not done permanently and people bring their cattle in these areas during better seasons. They migrate immediately when there is change of climate. These cattle are not only their food but also their wealth.

In Central Asia there are vast grasslands. In Europe, Britain, Switzerland, Holland and some areas of Denmark are most suitable for cattle rearing and establishing dairy farms. In New Zealand, Australia, areas of South Africa, South Canada and some areas of USA, cattle rearing is carried out on a large scale. For obtaining meat cattle rearing is also carried out in USA, South America (Argentina, Brazil), Africa (Sudan) Australia (Queens land) in Holland and England. These animals are slaughtered through machines and the meat is exported to other countries. Millions of people in different countries earn their living by this trade.

Further more, the introduction of cattle rearing to provide beef for export to markets in the United States, Japan, and Europe has resulted in widespread deforestation for creating grazing lands. Since the pasture is often of poor quality, large areas may be needed to support relatively few animals. As it is expensive to maintain grazing land of even this low quality, the easy option is to clear more forests rather than investing in fertilizer or reducing herd size to sustainable levels.

As the population of countries would expand, there would be scarcity of milk, butter and cheese. we would need to have more cattle rearing to avoid scarcity. The milk and its derived products are very essential to maintain our health. The children will face many difficulties if there would be shortage of milk. There fore, every goverment accross the world is adopting latest ways to rear cattles. An others solution to this is to control the population expansion. If parents would have fewer children, they would be in a position to nourish and educate their children in a pleasing manner. In this way, literate parents would take care of their children in a better way. Also, they would care for balanced diet of children.

Cultivation/Farming

Cultivation or farming is one of the ancient professions. It developed in ancient times as hunters and gatherers of the Stone Age turned to the cultivation of favoured varieties of crops. Modern crops were gradually derived from their wild ancestors through continual selection for larger seed size, improved fruit, and other desired traits.



Fumigation being done for pest control

THE WORLD POPULATION ENGAGED IN AGRICULTURE

